Basics of Mech. Engg (ME 101)

MOD: 2

Kinematics of Rigid Bodies

Theory Part →

Engg. Mechanics, 2nd ED. by Basudeb Bhattacharya

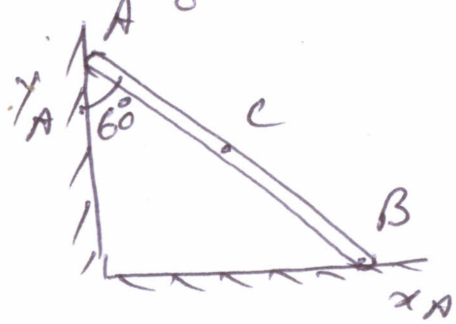
Ch-9, pp-510-516.

[9.2, 9.3, 9.4, 9.5, 9.6 & 9.7]

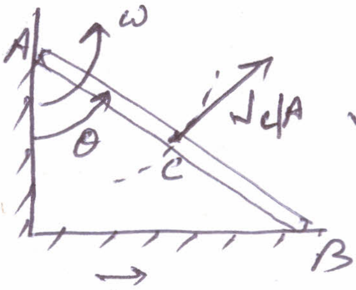
The examples are given in separate pages.
(Exps are from (1) B. Bhattacharya & (2) Rajasekharan
11 examples)

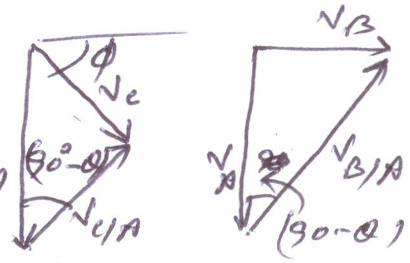
Kinematics of Rigid bodies

Example: A bar AB of length 5m slides in xy plane. The vel of pt. A shown is 10 m/s downwards and makes an angle 60° with vertical. Determine the vel of pt B & mid-point C.



Soln:





As Vel at pt C $\vec{V}_C = \vec{V}_A + \vec{V}_{C/A}$

in which $\vec{V}_A$ is known & $\vec{V}_{C/A}$ is $\perp^r$ to AB

$\therefore \vec{V}_{C/A} = \omega(AC) = 2.5\omega$

From geometry $y_A = 5\cos\theta$ or $\dot{y}_A = 5\dot{\theta}\sin\theta = -10$

$\therefore \omega = \dot{\theta} = \frac{2}{\sin 60^\circ} = 2.309$ m/s (Anticlock wise)

$\vec{V}_{C/A} = 2.309 \times 2.5 = 5.77$ m/s

$V_C \cos\phi = V_{C/A} \sin(90-\theta) = V_{CA}\cos\theta$

or $V_C \cos\phi = 5.77 \cos 60^\circ = 2.885$

As $V_C \sin\phi = -V_A + V_{C/A}\sin\theta = -10 + 5.77 \times \sin 60^\circ = -5$ m/s

$\therefore V_C = \sqrt{2.885^2 + 5^2} = 5.77$ m/s ; $\tan\phi = \frac{5}{2.885}$; $\phi = 60^\circ$

To find $\underline{V_B}$; from vector diagram $\vec{V}_B = \vec{V}_A + \vec{V}_{B/A}$

or $V_B^2 = V_{B/A}^2 - V_A^2$ or $V_{B/A} = 2V_{C/A} = 2 \times 5.77 = 11.54$ m/s

or $V_B^2 = (11.54)^2 - 100 = 5.75$ m/s $\longrightarrow$

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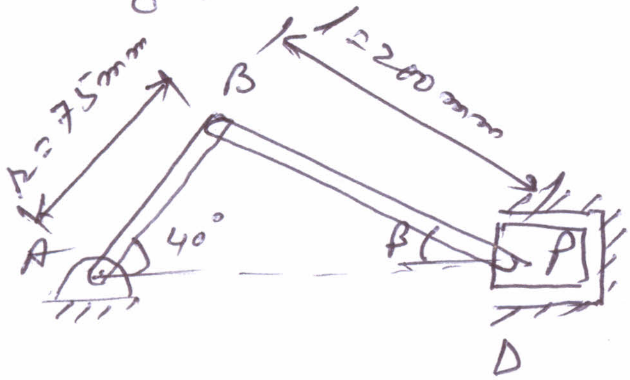
<u>Example</u> In the engine system shown in Fig., the crank AB has a constant angular vel of 4000 rpm. For the crank position indicated, find (a) the angular vel of connecting rod BD (b) the vel of piston P.



<u>Soln</u> Motion of crank: The crank AB rotates about a pt A

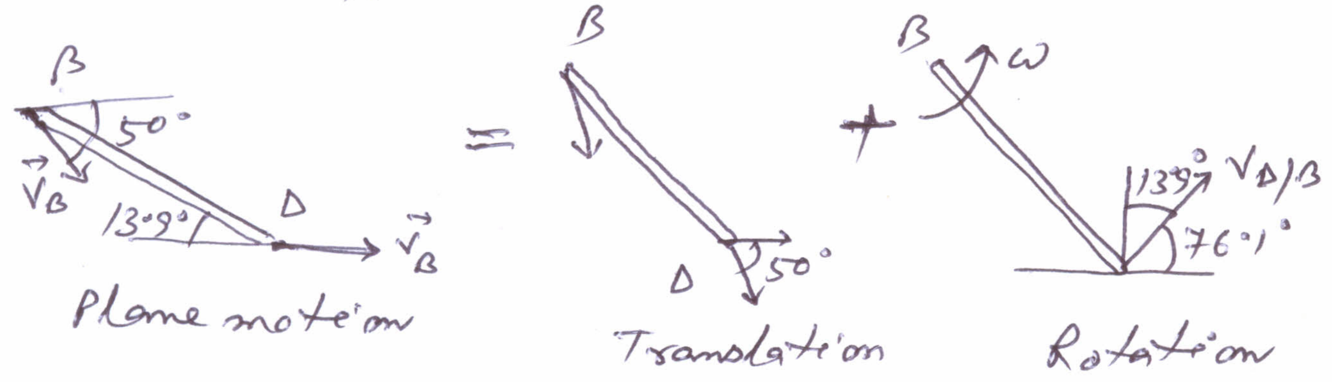
$$\omega_{AB} = \frac{4000}{60} \text{ rev/sec} = \frac{4000}{60} \times 2\pi = 418 \text{ rad/s}$$

$$V_B = \omega_{AB} \times r = 418 \times 0.075 = 31.36 \text{ m/s} \quad [V_B \text{ is } \perp^r \text{ to AB}]$$

Motion of connecting rod BD: From triangle ABD,

$75 \sin 40° = 200 \sin \beta \; \therefore \beta = 13.9°$

$\vec{V}_D = \vec{V}_B + \vec{V}_{D/B}$



PTO

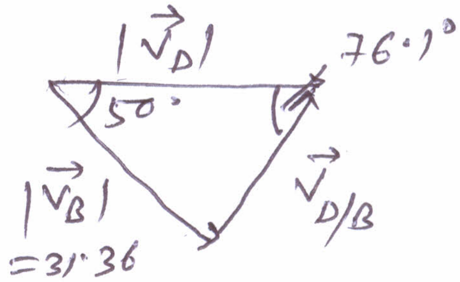
Drawing vel diagram

From triangle

$$\frac{|\vec{V}_D|}{\sin(180^\circ-(50^\circ+76.1^\circ))} = \frac{31.36}{\sin 76.1^\circ} = \frac{|\vec{V}_{D/B}|}{\sin 50^\circ}$$

$$|\vec{V}_{D/B}| = \frac{31.36 \times \sin 50^\circ}{\sin 76.1^\circ} = 24.74\ m/s$$

$$|\vec{V}_D| = \frac{31.36 \times \sin 53.9^\circ}{\sin 76.1^\circ} = 26\ m/s$$



Since $|\vec{V}_{B/D}| = 0.2\ \omega_{BD}$

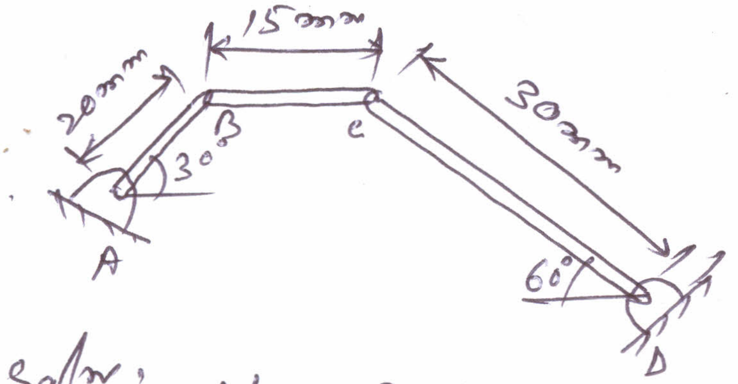
$\therefore 0.2\ \omega_{BD} = 24.74$

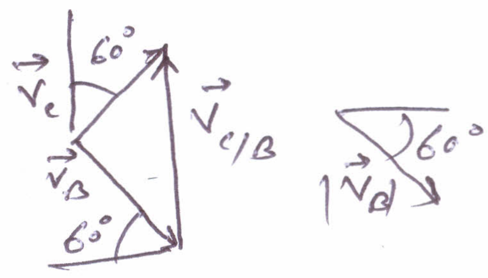
or $\omega_{BD} = \frac{24.74}{0.2} = 123.7$ Rad/sec

— a —

Kinematics of Rigid Bodies

Example: The linkage ABCD as shown in Fig moves in a vertical plane. At the instant shown, crank AB has a clockwise angular velocity of 8 rad/s. Determine the angular velocities of links BC & CD.





Soln:

$V_B = \omega_{AB}(AB) = 8 \times 20 = 160$ mm/s

$\vec{V}_C = \vec{V}_B + \vec{V}_{C/B}$. Magnitude of V_B & dir of V_C, V_B, $V_{C/B}$ are known.

From vector diagram

$|\vec{V}_B| \cos 60^\circ = (\vec{V}_C) \sin 60^\circ$; $160 \times \frac{1}{2} = |\vec{V}_C| \times 0.866$

$\therefore \vec{V}_C = \dfrac{80}{0.866} = 92.38$ mm/s

$|\vec{V}_B| \sin 60^\circ + |\vec{V}_C| \cos 60^\circ = |\vec{V}_{C/B}|$

or $160 \times 0.866 + 92.38 \times 0.5 = \boxed{184.75 = |\vec{V}_{C/B}|}$ mm/s

$\omega_{BC} = \dfrac{|\vec{V}_{C/B}|}{15} = \dfrac{184.75}{15} = 12.32$ rad/s (anticlockwise)

$\omega_{CD} = \dfrac{|\vec{V}_C|}{30} = \dfrac{92.38}{30} = 3.08$ rad/s (clockwise)

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Kinematics of Rigid Bodies

Example: A roller moves within two conveyor belts, of which top belt moves at the speed of 8 m/s and bottom belt moves at 6 m/s. Determine the linear and angular vel of the roller if the dia of the roller is 55 cm as shown in Fig 1.

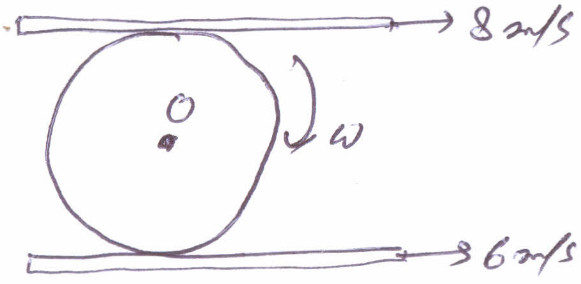


Fig 1.

Fig 2

Fig 3

Soln Ist Approach

Let the vel & angular vel of centre O be v_0 & ω. vel of top & bottom plates are v_{p_1} & v_{p_2}. velocities of each point on the circumference, w.r.t 'O' is constant = rω acting tangentially. Now vel at pt A [Fig 2]

$$v_A = v_0 + v_{A/0} \quad \therefore \; 8 = r\omega + v_0 \quad \text{--- (1)}$$

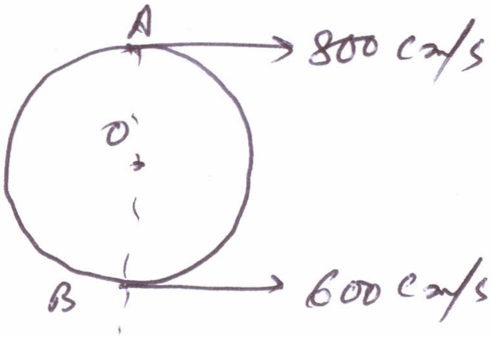
Similarly vel at pt B [Fig 3]

$$v_B = v_0 + v_{B/0} \quad \therefore \; 6 = -r\omega + v_0 \quad \text{--- (2)}$$

Solving (1) & (2) & substituting $r = \frac{.55}{2}$ m we get $v_0 = 7$ m/s & $\omega = 3.636$ rad/sec Ans

Contd to next page (PTO) for 2nd approach

2nd APPROACH



Dia of roller = 55 cm

Let $r = \vec{BA} = 55\hat{j}$,

vel at A, $v_A = 800\hat{i}$,

" B $v_B = 600\hat{i}$

& $\vec{\omega}$ = Angular vel of roller $= \omega\hat{k}$

Now $\vec{v}_A = \vec{v}_B + \vec{\omega} \times r$

or $800\hat{i} = 600\hat{i} + \omega\hat{k} \times 55\hat{j}$

or $200\hat{i} = -55\omega\hat{i}$

or $\omega = -3.636$

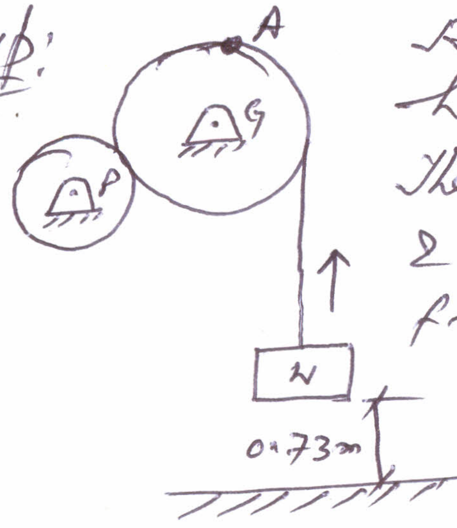
So, angular vel of roller = 3.636 rad/s (clockwise)

Again, $\vec{v}_0 = \vec{v}_B - 3.636\hat{k} \times \frac{55}{2}\hat{j}$

$= 600\hat{i} + 100\hat{i} = 700\hat{i}$

So linear vel of roller = 700 cm/s = 7 m/s

Kinematics of Rigid Bodies



Exp: A pinion P & gear G are attached for hoisting a load W, as shown in Fig. The radii of pinion & gear are 95 mm & 350 mm, respectively. The load is lifted from anywhere & after passing a distance 0.73 m acquires a vel of 2.1 m/s, with constant acc. Determine (i) The acc of pt 'A' on the cable in contact with gear & (ii) Angular speed & angular ~~vel~~ acc of pt 'P'.

Soln: To determine the upward acc of the block

$$v^2 = 0^2 + 2as \quad \text{or} \quad a = \frac{v^2}{2s} = \frac{2.1^2}{2\times 0.73} = 3.02 \text{ m/s}^2$$

So, normal acc $a_n = \frac{v^2}{r_G} = \frac{2.1^2}{0.35} = 12.6 \text{ m/s}^2$

Tangential acc $a_t = a = 3.02 \text{ m/s}^2$

At pt 'A' the total acc

$$a_A = \sqrt{a_n^2 + a_t^2} = \sqrt{12.6^2 + 3.02^2} = 12.957 \text{ m/s}^2$$

Now angular speed of gear G, $\omega_G = \frac{v}{r_G} = \frac{2.1}{0.35} = 6 \text{ rad/s}$

Angular acc of gear G, $\alpha_G = \frac{a_t}{r_G} = \frac{3.02}{0.35} = 8.63 \text{ rad/s}^2$

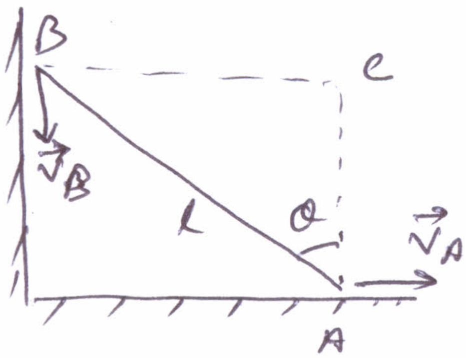
From condition of compatibility, at the pt of contact of pinion & gear, $\omega_p r_p = \omega_G r_G$ —(1) & $\alpha_p r_p = \alpha_G r_G$ —(2)

Using (1) $\omega_p = \omega_G \times \frac{r_G}{r_p} = 6 \times \frac{0.350}{0.095} = 22.105 \text{ rad/s}$

Using (2) $\alpha_p = \alpha_G \times \frac{r_G}{r_p} = 8.63 \times \frac{0.350}{0.095} = 31.795 \text{ rad/s}^2$

concept of Instantaneous centre w.r.t. ladder prob.

A ladder shown in Fig moves horizontally at A & vertically at B. Since the vel at A is horizontal, the instantanious centre of rotation must lie on the vertical line through A, & since the vel at B is vertical, the instantanious centre will ~~be~~ be on horizontal line through B. Hence it is the intersection pt 'C'.



Knowing $|\vec{V}_A|$ one can calculate ω (angular vel)

as $$\omega = \frac{|\vec{V}_A|}{l\cos\theta} = \frac{V_A}{l\cos\theta}$$

Hence, $|\vec{V}_B| = \omega \times BC$

$$|\vec{V}_B| = \frac{|\vec{V}_A|}{l\cos\theta}\, l\sin\theta = V_A\tan\theta$$

Note: Only abs vel are involved in the computations.

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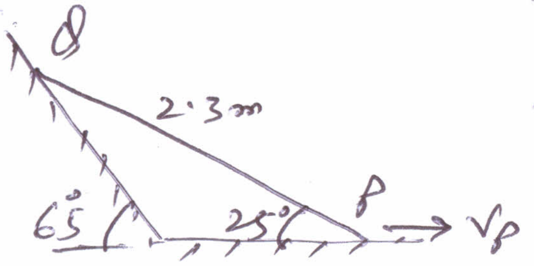
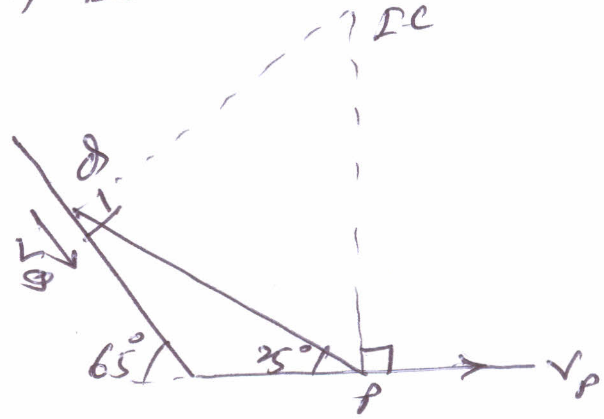
Kinematics of Rigid Bodies (I.C. Method)

Example: In the configuration of the solid bar PQ as shown in fig, with $v_P = 2.4$ m/s constant, determine the angular speed of PQ & the vel of end Q.



Soln: The vel diagram with I.C. of rotation is shown below.



From geometry

$\angle OQP = 65^\circ - 25^\circ = 40^\circ$

$\angle IC\,QP = 90^\circ - 40^\circ = 50^\circ$

$\angle IC\,PQ = 90^\circ - 25^\circ = 65^\circ$

& $\angle P\,IC\,Q = 180^\circ - (50^\circ + 65^\circ) = 65^\circ$

Now applying Sine rule in $\Delta P\,IC\,Q$

$$\frac{PQ}{\sin\angle P\,IC\,Q} = \frac{IC\,P}{\sin\angle IC\,QP} = \frac{IC\,Q}{\sin\angle IC\,PQ}$$

or $\dfrac{2.3}{\sin 65^\circ} = \dfrac{IC\,P}{\sin 50^\circ} = \dfrac{IC\,Q}{\sin 65^\circ}$ $\Rightarrow$ so $IC\,Q = 2.3$ m

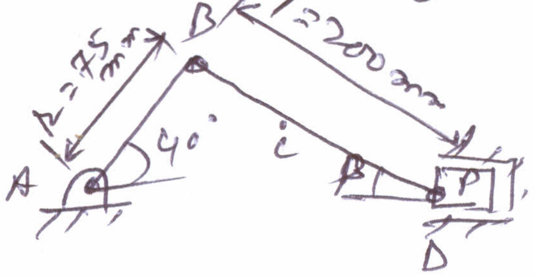
& $IC\,P = 1.944$ m

Again $\omega \times IC\,P = v_P$ or $\omega = \dfrac{v_P}{IC\,P} = \dfrac{2.4}{1.944} = 1.235$ rad/s

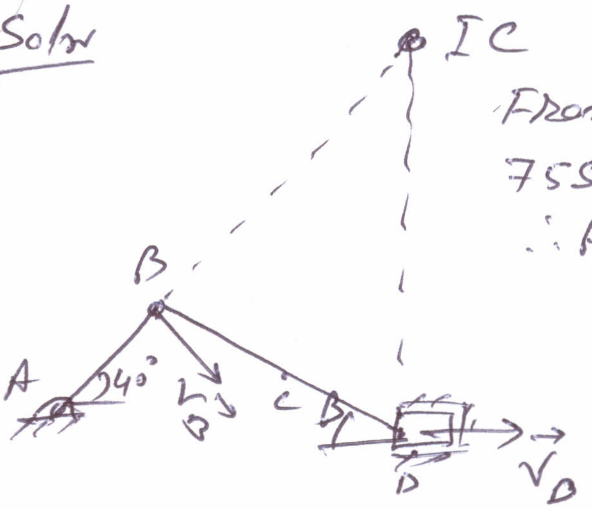
So, $v_Q = \omega \times IC\,Q = 1.235 \times 2.3 = 2.84$ m/s

Instantanious Centre

Example: In the engine sys shown in Fig.. The crank AB has a constant angular vel of 4000 rpm. For the crank position indicated, determine (a) Angular vel of rod BD, (b) vel of piston P. [use Instantanious Centre method to solve it]



Soln



From $\triangle ABD$

$75 \sin 40° = 200 \sin \beta$

$\therefore \beta = 13.9°$

$\omega_{AB} = \frac{4000}{60} \times 2\pi$

$= 418$ rad/s

$v_B = 418 \times \frac{75}{1000} = 31.36$ m/s

I.C. can be found by drawing $\perp^r$ to $\vec{v}_B$ at B & $\vec{v}_D$ at D.

$(B - IC)\cos 40° = 200 \cos 13.9° \Rightarrow \quad B - I.C = \frac{194.14}{0.766} = .253$ mm

$(D - IC) = (A - IC) \sin 40°$

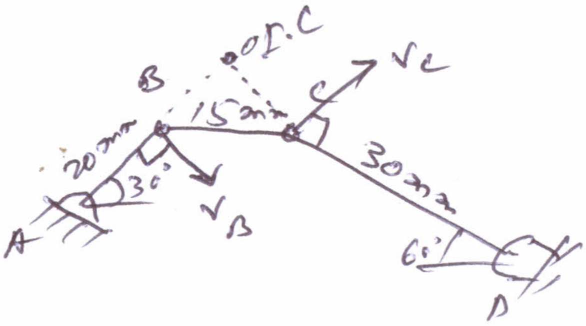
$= (75 + 253.44) \times 0.642 = 0.210$ mm

$\omega_{(IC-B)} = \frac{v_B}{B - IC} = \frac{31.36}{0.253} = 123.95$ rad/s

$v_D = \omega_{(IC-B)} \times (IC - D) = 123.95 \times 0.210 = 26$ m/s

— x —

The linkage ABCD shown in Fig moves in the vertical plane. At the instant shown crank AB has a clockwise angular vel of 8 rad/s. Find angular velocities of links BC & CD using I.C. method:



As $B-IC = BC\cos 30° = 15 \times 0.866$
$= 12.990$ mm

$A-IC = 20 + 12.99 = 32.99$ mm

$V_B = \omega \times 20 = 8 \times 20 = 160$ mm/s

$$\omega_{BC} = \frac{160}{12.99} = 12.32 \text{ rad/s (Anticlockwise)}$$

$V_c = (IC - C) \times \omega_{BC} = 12.32 \times BC \sin 30°$

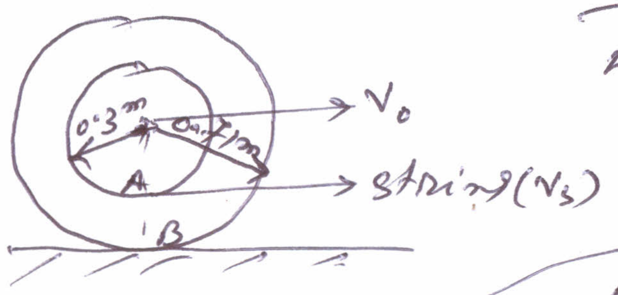
$= 12.32 \times 15 \times \frac{1}{2} = 92.38$ m/s

Angular vel to link $CD = V_c/30$

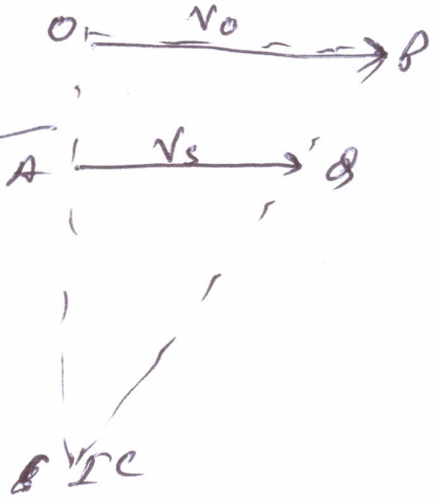
or $$\omega_{CD} = \frac{92.38}{30} = 3.076 \text{ rad/s.}$$ Ans

Instantanious Centre Method

Example: A wheel is made to roll without slipping towards the right, by pulling a string wrapped around a coaxial spool as shown in Fig. Determine the velocity with which the string should be pulled so that the centre of the wheel moves with a vel of 13.5 m/s



Soln: By drawing the vel diagram with I.C. of Rotation.



From geometry,

$OA = 0.3\,m$, $OIC = 0.71\,m$

Using similar triangles

$\Delta\, IC\,OP$ & $\Delta\, IC\,AQ$,

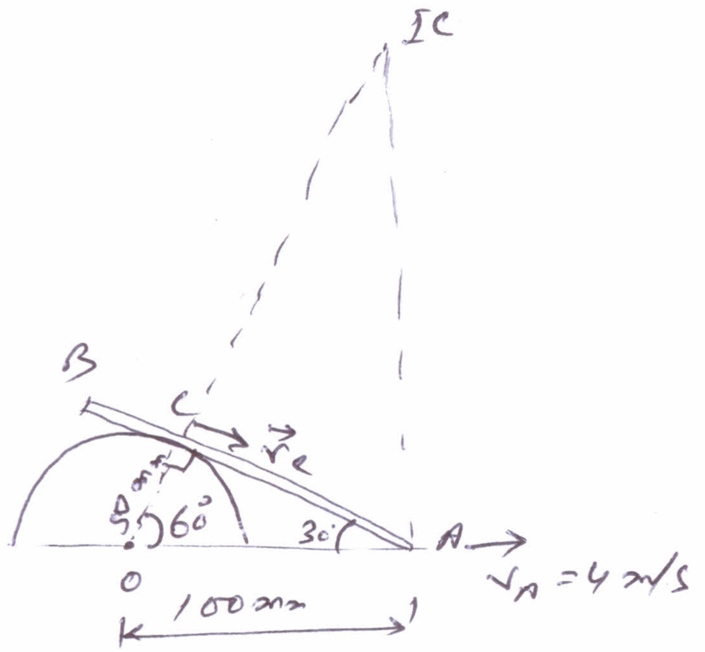
$$\frac{V_0}{OIC} = \frac{V_s}{AIC} = \frac{V_s}{OIC - OA}$$

So we get

$$\frac{13.5}{0.71} = \frac{V_s}{0.71 - 0.3} \quad \Rightarrow \quad V_s = 7.796\ m/s$$

Instantaneous Centre Method

Example: The end 'A' of a bar AB moves with a constant vel, $V_A = 4$ m/s along the horizontal surface such that it always remains in contact with a disc of radius $r = 50$ mm which is resting on the horizontal surface as shown in fig. Find the angular vel of the bar at an instant when the bar makes an angle $\alpha = 30°$ with the horizontal.



Soln $OA = 100$ mm

$(O - IC)\cos 60° = 100 \text{ mm} = 0.1 \text{ m}$

$\therefore (O - IC) = 0.2$ m.

$(IC - A) = 200 \sin 60° = 200 \times .866 = 173.2 \text{ mm} = 0.1732 \text{ m}$

$$\omega_{(IC-A)} = \frac{4}{0.1732} = 23 \text{ rad/s}$$

$\therefore$ The angular vel of the bar $= 23$ rad/s Ans

Relative motion of two particles/Bodies

In relative motion, the concept of translating co-ordinate system is used where as in (abs) motion, the co-ordinates are fixed.

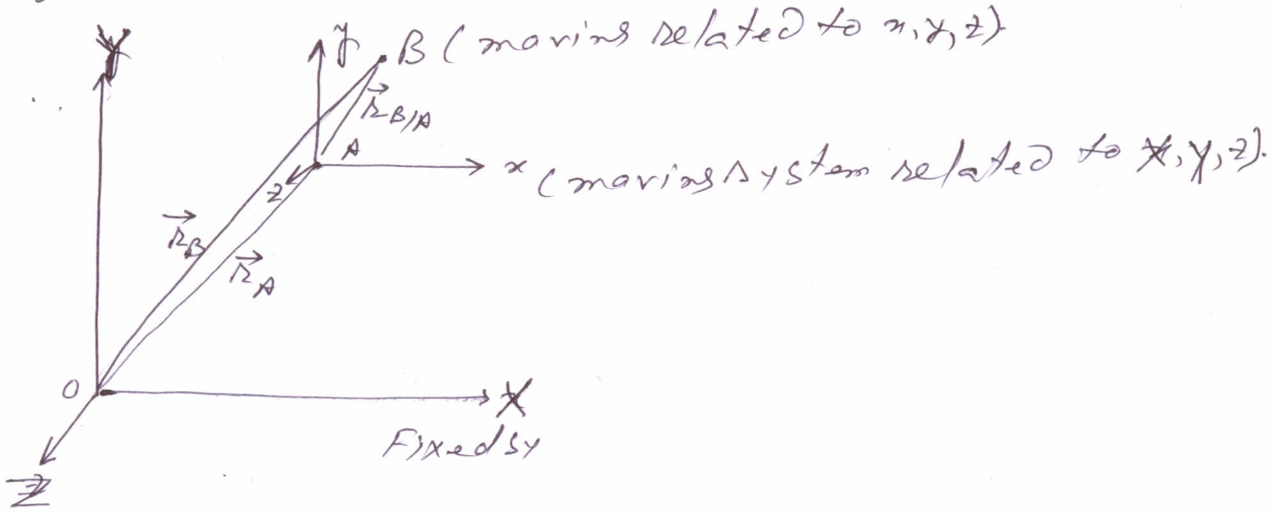


Fig: Translation of xyz w.r.t. XYZ

$$\vec{r}_B = \vec{r}_A + \vec{r}_{B/A} \quad \text{--- (1)}$$

$$\vec{v}_B = \dot{\vec{r}}_B = \dot{\vec{r}}_A + \dot{\vec{r}}_{B/A} = \vec{v}_A + \vec{v}_{B/A} \quad \text{--- (2)}$$

$$\vec{a}_B = \ddot{\vec{r}}_B = \ddot{\vec{r}}_A + \ddot{\vec{r}}_{B/A} = \vec{a}_A + \vec{a}_{B/A} \quad \text{--- (3)}$$

The above relations are used to determine

(i) position of two particles moving in diff dir after 't' sec.

(ii) The distance between the two particles after 't' sec.

(iii) The time taken for the travel if the distance between them 'd' is known

(iv) When two particles move in different directions from two different positions, min^{m} distance between them

(v)

Prob on relative motion of two particles/Bodies

Q. The position vector of particles/Bodies A & B are defined in fixed XY frame by $\vec{r}_A = -4.5t^3\hat{i} - 1.5t^3\hat{j}$ & $\vec{r}_B = -6t^2\hat{i} + 2.25t^3\hat{j}$ where $\vec{r}_A$ & $\vec{r}_B$ are in metres and & t is in sec. Determine $\vec{V}_{B/A}$ & $\vec{a}_{B/A}$ after 3 sec and also $\vec{V}_B$, $\vec{a}_B$.

Soln: $\vec{r}_A = -4.5t^3\hat{i} - 1.5t^3\hat{j}$, $\vec{r}_B = -6t^2\hat{i} + 2.25t^3\hat{j}$

$$\vec{r}_{B/A} = \vec{r}_B - \vec{r}_A = [-6t^2 + 4.5t^3]\hat{i} + [3.75t^3]\hat{j}$$

$$\vec{V}_{B/A} = \frac{d\vec{r}_{B/A}}{dt} = [-12t + 13.5t^2]\hat{i} + [11.25t^2]\hat{j}$$

$$\vec{a}_{B/A} = \frac{d\vec{V}_{B/A}}{dt} = [-12 + 27t]\hat{i} + [22.5t]\hat{j}$$

when $t = 3$ sec,

$$\vec{r}_{B/A} = 67.5\hat{i} + 101.25\hat{j}$$

$$\vec{V}_{B/A} = 85.5\hat{i} + 101.25\hat{j}$$

$$\vec{a}_{B/A} = 69\hat{i} + 67.5\hat{j}$$

~~at t = 3 sec,~~

~~$\vec{r}_{B/A}$~~

At $t = 3$ sec,

$$\vec{r}_A = -121.5\hat{i} - 40.5\hat{j}$$

$$\vec{V}_A = -121.5\hat{i} - 40.5\hat{j}$$

$$\vec{a}_A = -81\hat{i} - 27\hat{j}$$

$$\therefore \vec{r}_B = \vec{r}_A + \vec{r}_{B/A} = -54\hat{i} + 60.75\hat{j}$$

$$\vec{V}_B = \vec{V}_A + \vec{V}_{B/A} = -36\hat{i} + 60.75\hat{j}$$

$$\vec{a}_B = \vec{a}_A + \vec{a}_{B/A} = -12\hat{i} + 40.5\hat{j}$$

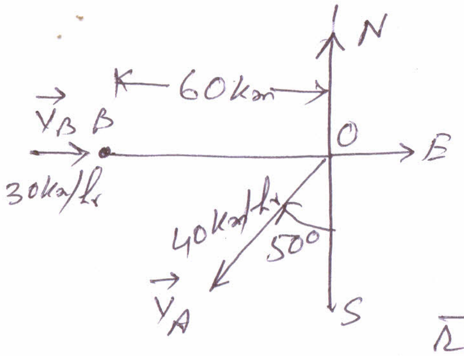
& The distance between A & B after $t = 3$ sec

$$\vec{r}_{B/A} = 67.5\hat{i} + 101.25\hat{j} = 121.687\text{ m} \quad \text{Ans}$$

— x —

Problem of relative motion of two particles/Bodies

Q. Ship B is approaching a port, moving in East direction with a speed of 30 km/hr. When it is at 60 km away from the port, another ship 'A' leaves the port with a speed of 40 km/hr in S 50° W. If they can exchange signals when they are within 35 km distance, find when & how long of they can exchange the signals.



Soln: $\vec{V}_B = 30\hat{i}$

$\vec{V}_A = -40\sin 50°\hat{i} - 40\cos 50\hat{j} = -30.642\hat{i} - 25.712\hat{j}$

$\vec{R}_B = -60\hat{i} + (30)t\,\hat{i}$,

$\vec{R}_A = -(30.642)t\,\hat{i} - (25.712t)\hat{j}$

$\vec{R}_{A/B} = \vec{R}_A - \vec{R}_B = (60 - 30.642t)\hat{i} - (25.712t)\hat{j}$

$|\vec{R}_{A/B}|$ = distance between A & B

$= \left[(60 - 30.642t)^2 + (-25.712t)^2\right]^{\frac{1}{2}}$

They can exchange signals when distance between them is less than ≤ 35 km.

So $(60 - 30.642t)^2 + (-25.712t)^2 = 35^2$ $\Rightarrow t = ??$

or $t^2 - 1.677t + 0.547 = 0$

or $t = \left\{1.677 \pm \sqrt{(1.677)^2 - 4(0.547)^2}\right\}/2$

$\therefore t = 0.444$ & $t = 1.233$ hr

When $t = 0.444$ hr $= 26$ min 38 sec

When $t = 1.233 = 1$ hr 14 min

They can exchange signals from 26 min 38 sec to 1 hr 14 min

$\therefore \Delta t = 1.233 - 0.44 = 0.789$ hr $= 47$ min 22 sec. Ans

— x —

MOD-2 ~~PART B~~

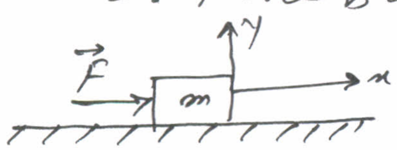
Newtons Law for rectangular co-ordinates:

In rectangular co-ordinates, one can express Newton's Law as follows:

$$\left.\begin{aligned} F_x &= ma_x = m\frac{dv_x}{dt} = m\frac{d^2x}{dt^2} = m\ddot{x} \\ F_y &= ma_y = m\frac{dv_y}{dt} = m\frac{d^2y}{dt^2} = m\ddot{y} \\ F_z &= ma_z = m\frac{dv_z}{dt} = m\frac{d^2z}{dt^2} = m\ddot{z} \end{aligned}\right] \quad -(1)$$

Rectilinear Translation:

Case-I: Force is a function of time or a constant



A particle of mass m acted on by a time varying force F(t) is shown in fig (frictionless surface). The force of gravity is equal & opp. to the normal force from the plane so that F(t) is the resultant force acting on the mass. The Newton's law can be written as:

$$F(t) = m\frac{d^2x}{dt^2} \quad \therefore \frac{d^2x}{dt^2} = \frac{F(t)}{m} \quad -(2)$$

Knowing the acc in x dir, we can solve for F(t).

The inverse prob, when F(t) is known & wish to determine the motion, requires integration. The eqn (2) can be written as

$$\frac{d}{dt}\left(\frac{dx}{dt}\right) = \frac{F(t)}{m} \quad \text{or} \quad d\left(\frac{dx}{dt}\right) = \frac{F(t)}{m}dt$$

integrating both sides we get: $\frac{dx}{dt} = v = \int \frac{F(t)}{m}dt + C_1 \quad -(3)$

or $x = \int\left[\int \frac{F(t)}{m}dt\right]dt + C_1 t + C_2 \quad -(4)$

when $t=0$, $v = v_0$ & $x = x_0$ $-(5)$. So C_1 & C_2 can be calculated

Case-II: Next page

Case-II : Force is a function of speed:

Here in this case where the resultant force on the particle depends only on the value of speed of the particle. An exp. of such a force is the aerodynamic drag force on an airplane or missile. So Newton's law can be written in the following:

$$\frac{dv}{dt} = \frac{F(v)}{m} \quad —(6)$$

Rearranging eqn (6) $\frac{dv}{F(v)} = \frac{1}{m}dt.$

or $\int \frac{dv}{F(V)} = \frac{1}{m}t + C_1 \quad —(7)$. The result will give t as a function of V. However, generally it is preferred to solve v in terms of t, such that

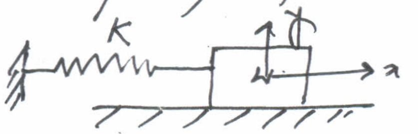
$V = H(t, C_1)$. Where H is a function of t & constant of integration C_1. A second integration may be performed by replacing v by $\frac{dx}{dt}$ & bringing dt to RHS of the eqn. such that

$$x = \int H(t, C_1)\,dt + C_2 \quad —(8)$$

The values of C_1 & C_2 can be found from initial conditions.

Case-III : Force is a function of position

A simplest example of such case is the frictionless mass-spring system.



The horizontal force from the spring at all positions of the body will be function of position x.

The Newtons law for position-dependent forces can be written as:

$$m\frac{dv}{dt} = F(x) \quad —(9)$$

See Next Page

Case-III Contd.

Eqn. (9) can be written as:

$$m\frac{dv}{dt} = m\frac{dv}{dx}\frac{dx}{dt} = mv\frac{dv}{dx}$$

Now separating the variables in eqn. (9) as follows:

$$mv\,dv = F(x)\,dx$$

integrating, we get $\frac{mv^2}{2} = \int F(x)\,dx + C_1$ — (10)

Solving for v & using $\frac{dx}{dt}$ in its place, we get

$$\frac{dx}{dt} = \left[\frac{2}{m}\int F(x)\,dx + C_1\right]^{\frac{1}{2}}$$

again separating the variables & integrating again, we get

$$t = \int \frac{dx}{\left[\frac{2}{m}\int F(x)\,dx + C_1\right]^{\frac{1}{2}}} + C_2 \quad \text{— (11)}$$

For a given F(x), v & x can be evaluated as a function of time from eqn. (10) & (11). The constants of integration can be found out from initial conditions.

The force from a spring will be proportional to x measured from position of W corresponding to undeformed configuration of the system. So the force will have a magnitude |kx|, where k is the spring constant, is the force needed on the spring per unit elongation or compression of spring. When x has a (+)ive value, the spring force points in (-)ive direction. The spring force is for reason called a restoring force & expressed as −kx.

Q. Prob. on force is a function of time. [Case - I]

A 100N body is initially stationary on a 45° incline as shown in Fig. The co-efficient of dynamic friction μ_d between the block & incline is 0·5. What distance along the incline must the weight slide before it reaches a speed of 40 m/s.?

Soln

Fig Body slides on incline

FBD

For equilibrium

$$100 \cos 45° = N = 70{\cdot}71 \text{ Newton} \quad \text{—(a)}$$

Applying Newton's Law along incline we have

$$\frac{100}{g}\frac{d^2s}{dt^2} = 100 \sin 45° - \mu_d N \quad \text{—(a)}$$

$$\therefore \frac{d^2s}{dt^2} = 3{\cdot}46 \quad \text{—(b)}$$

Rewriting (b) $d\left(\frac{ds}{dt}\right) = 3{\cdot}46\,dt$ & integrating, we have

$$\frac{ds}{dt} = 3{\cdot}46\,t + C_1 \quad \text{—(c)} \quad \&$$

$$s = 3{\cdot}46\,\frac{t^2}{2} + C_1 t + C_2 \quad \text{—(d)}$$

When $t = 0$, $s = \frac{ds}{dt} = 0$, so, $C_1 = C_2 = 0$. When $\frac{ds}{dt} = 40$ m/s, we have for t from ~~eq~~ⁿ (c)

$$40 = 3{\cdot}46\,t \quad \therefore t = 11{\cdot}56 \text{ sec}$$

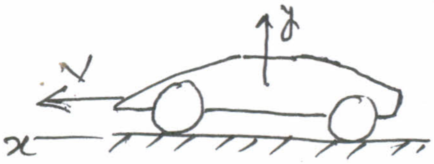
Substituting value of t in eqⁿ (d), we can get distance travelled to reach the speed of 40 m/s as follows

$$s = \frac{3{\cdot}46\,(11{\cdot}56)^2}{2} = \underline{231{\cdot}18 \text{ m}} \quad \text{Ans}$$

— x —

Q: Motion on force is a function of speed. [Case-II]

A high speed car is moving at a speed of 100 m/s. The resistance to motion of the vehicle is mainly due to aerodynamic drag, which for this speed can be approximated as $0.2v^2$ N with v in m/s. If the vehicle has a mass of 4000 kg, what distance will it move before its speed is reduced to 70 m/s?



Soln

We have using Newtons Law in this case

$$\frac{dv}{dt} = \frac{-F(v)}{m} \quad \text{—}$$

$-F(v)$ means drag force opposing motion.

$$\text{or } \frac{dv}{dt} = \frac{-0.2v^2}{4000} = -5\times10^{-5}v^2 \quad \text{— (a)}$$

separating the variables, we get $\frac{dv}{v^2} = -5\times10^{-5}dt$ — (b)

Integrating we get $-\frac{1}{v} = -5\times10^{-5}t + C_1$ — (c). Taking $t=0$, when $v=100$, we get $C_1 = -1/1000$. Replacing v by $\frac{dx}{dt}$, we have next

$\frac{1}{v} = \frac{dt}{dx} = 5\times10^{-5}t + \frac{1}{100}$ — (d). Again separating the variables we get

$\frac{dt}{5\times10^{-5}t + (1/1000)} = dx$. Now let $z = 5\times10^{-5}t + (1/1000)$,

so $dz = 5\times10^{-5}dt$, so $\frac{dz}{z} = 5\times10^{-5}dx$.

Now integrating & replacing z, we get

$$\ln\left(5\times10^{-5}t + \frac{1}{100}\right) = 5\times10^{-5}x + C_2$$

when $t=0$, $x=0$, & so $C_2 = \ln(1/100)$.

so $\ln(5\times10^{-3}t + 1) = 5\times10^{-5}x$ — (e)

Substituting $v = 70$ in eqn (d), solve for t, we get $t = 85.7$ sec.

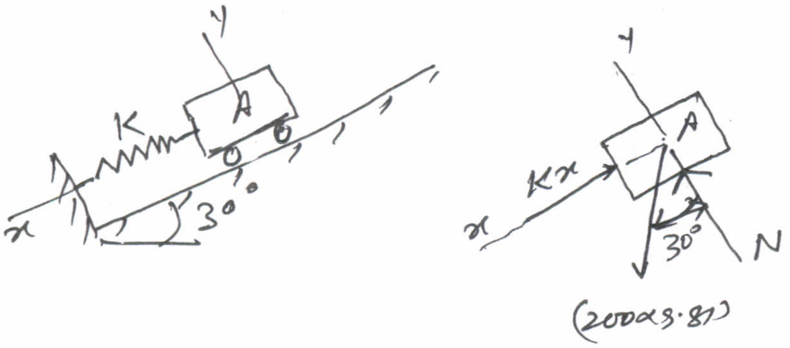
Finally to find x using eqn (e)

$$\ln[(5\times10^{-3})(85.7) + 1] = 5\times10^{-5}x$$

$\therefore$ $x = 7.13$ km Ans

The distance travelled is 7.13 km

Q. Prob. on force is a function of position [Case - III]
A cart as shown in Fig. having a mass of 200 kg is held on an incline so as to just touch an undeformed spring whose spring constant K is 50 N/mm (50×1000) N/m. If a body A is released very slowly, what distance down the incline A must move to reach an equilibrium position? If body A is released suddenly, what is the speed when it reaches the equilibrium condition which was prevailed at slow release?



FBD

Let δ represents compression of spring in mm. So

$$K = \frac{F}{\delta} \quad \therefore \delta = \frac{F}{K} = \frac{(200)(9.81)\sin 30}{50} = 19.62\text{ mm}$$

Thus spring will compressed 0.01926 m by the cart if it is allowed to move down the incline very slowly.

For the case of quick release, using Newton's law $m\frac{dv}{dt} = F(x)$.
Thus using x in meters so that K is (50×1000) N/m.

$$\therefore 200\ddot{x} = (200)(9.81)\sin 30 - (50)(1000)(x)$$

$$\therefore \ddot{x} = 4.905 - 250x$$

Rewriting $\ddot{x}$, we have $v\frac{dv}{dx} = 4.905 - 250x$

Separating the variables & integrating,

$$\frac{v^2}{2} = 4.905x - 125x^2 + C_1$$

Now at $x = 0$, when $v = 0$. so $C_1 = 0$

Final step, set $x = 0.01962$ mm (value of δ) & solve for v

$$V = \{2[(4.905)(0.01962) - (125)(0.01962)^2]\}^{\frac{1}{2}}$$

$$\therefore \underline{V = 0.310\text{ m/s}} \quad \text{Ans}$$